From Numerical Baselines to Gradient-Boosted Trees for Heart-Disease Classification

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***Abstract*—We classify self-reported heart disease in the 2015 BRFSS survey using six numerical methods and a NumPy gradient-boosted tree ensemble. The selected GBDT improves validation F1 from 0.4386 for logistic regression to 0.4462. On the test set, logistic regression scores 0.449 F1 and GBDT scores 0.454.**

I. Introduction

We classify self-reported coronary heart disease or heart attack (`_MICHD`) in the 2015 Behavioral Risk Factor Surveillance System (BRFSS) [1]. The course release contains 328,135 labeled respondents, 109,379 unlabeled test respondents, and 321 predictors covering demographics, behavior, health conditions, and healthcare use.

This dataset mixes numeric quantities, categorical answer codes, and structured missingness. For example, `PHYSHLTH=88` means zero days, while 77 and 99 mean an unknown or refused answer. Codebook-based rules drop 14 survey-design variables, two duplicate variables, and 14 raw mixed-unit variables with computed replacements; they also convert units and fill numeric values implied by skipped questions. This leaves 56 numeric and 235 categorical predictors. Unknown and refused categorical answers remain separate categories. Only 8.83% of training labels are positive.

We compare numerical baselines with neural networks and gradient-boosted decision trees (GBDT), which capture nonlinear interactions and learn how to route missing answers [2], [3].

II. Methods

A. Linear baselines

We implement the six methods required by the project as linear baselines. Let X include an intercept, let $y \in \{0,1\}^N$, and write $z = Xw$. The methods minimize either squared error or logistic loss:

$$L_{\text{MSE}}(w) = \frac{\|Xw - y\|_2^2}{2N}, \tag{1}$$

$$L_{\log}(w) = \frac{1}{N}\sum_{i=1}^{N}\left[\log(1 + e^{z_i}) - y_i z_i\right]. \tag{2}$$

Gradient descent (GD) updates $w \leftarrow w - \gamma \nabla L(w)$; stochastic GD (SGD) uses one sampled row per update. Least squares uses a pseudoinverse of $X^\top X$, which is singular for our one-hot design; ridge adds a regularizer $\lambda\|w\|_2^2$ and solves $(X^\top X + 2N\lambda I)w = X^\top y$. The other two methods minimize logistic loss by GD, with or without the same L_2 penalty.

In the experiments, MSE GD uses 100 steps at $\gamma = 0.02$ and SGD uses 10^6 single-row steps at $\gamma = 5 \cdot 10^{-6}$. Both logistic methods use 300 steps at $\gamma = 0.1$; ridge and regularized logistic regression use $\lambda = 10^{-3}$.

B. Neural networks

As a capacity check, we average five NumPy MLPs with ReLU widths 512/256, dropout 0.3, batch size 512, and AdamW updates [4] (initial rate 10^{-3}, weight decay 10^{-5}). Training runs for two epochs with cosine rate decay, selecting checkpoints on an inner 10% holdout. Numeric inputs use 16-bin piecewise-linear encoding; categorical inputs are one-hot encoded.

C. Gradient-Boosted Decision Trees (GBDT)

Starting from training log-odds, GBDT adds trees $F_{t+1}(x) = F_t(x) + \eta f_t(x)$ to reduce logistic loss [5]. Our NumPy implementation uses gradients and Hessians for splits and leaf values [2], with up to 64 bins per variable for histogram search [3]. A separate missing bin is routed to maximize loss reduction. We use depth 5, 550 trees, $\eta = 0.05$, leaf L_2 penalty 10, minimum child Hessian 5, and row/feature fractions 0.7/0.3; logits from seeds 0–2 are averaged.

D. Data processing

Each fold fits preprocessing on training rows and applies it to validation rows. Linear models fill missing numeric values with training medians, add missingness flags, one-hot encode categories seen at least 30 times, and standardize the resulting $\sim$ 1,530 columns. GBDT uses the 291 cleaned variables directly.

E. Validation

All local evaluations split the labeled training set; the organizers withhold test labels and score uploaded predictions on AIcrowd. Predicting all negatives gives 91.17% accuracy but zero F1, motivating positive-class F1 and threshold-independent ROC AUC. The numerical baselines use five shuffled folds (seed 42), with each decision threshold fitted

Table I
NUMERICAL BASELINES ON FIVE VALIDATION FOLDS (SEED 42). F1 IS MEAN ± FOLD SD; ROC AUC IS THE FOLD MEAN. ONLY THE RAW LEAST-SQUARES BASELINE USES UNENCODED INPUTS.

Method	F1 (mean ± SD)	AUC
Raw inputs / LS	0.4095 ± 0.0019	0.845
MSE / GD	0.4289 ± 0.0022	0.858
MSE / SGD	0.4255 ± 0.0033	0.855
Least squares	0.4301 ± 0.0018	0.857
Ridge	0.4297 ± 0.0031	0.858
Logistic / GD	**0.4390** ± 0.0027	0.864
L2 logistic / GD	0.4388 ± 0.0022	0.864

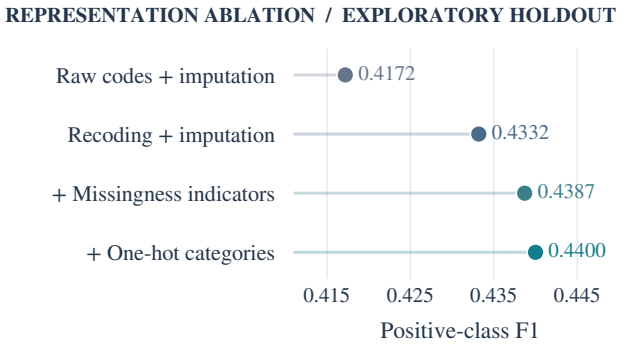


Figure 1. Exploratory logistic-regression F1 on one 80/20 holdout. The threshold was optimized on the holdout, making F1 optimistic. Recoding also removes survey-design fields and changes categorical unknown codes, unlike the final pipeline.

Table II
MODEL COMPARISON ON 15 VALIDATION FOLDS (FIVE FOLDS WITH THREE SPLIT SEEDS) AND ON SUBMITTED TEST PREDICTIONS. CV THRESHOLDS COME FROM OTHER FOLDS. BOLD MARKS THE FINAL MODEL; DASHES MEAN NO REPORTED SCORE.

Model	CV F1	CV AUC	Test F1
Logistic regression	0.4386	0.8643	0.449
Five-network MLP	0.4399	0.8655	–
Single GBDT (9-run mean)	0.4457	0.8698	–
Three-seed GBDT	**0.4462**	**0.8703**	**0.454**
GBDT + logistic blend	0.4463	0.8703	–
Seven-config GBDT bag	0.4469	–	–
GBDT + neural-net blend	0.4469	–	–
Feature GBDT ensembles	0.4474	–	–

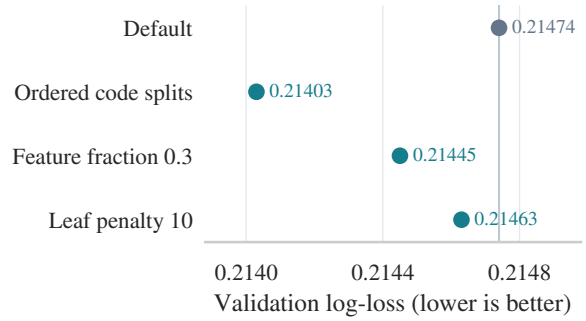


Figure 2. One-factor GBDT tuning on five validation folds (seed 42). Each point is mean log-loss at its best tree count. Changes are evaluated separately, not cumulatively.

on training scores. The GBDT comparison repeats five folds with split seeds 42–44; each scored fold uses a threshold chosen from the other folds' out-of-fold predictions.

III. RESULTS

Representation and linear baselines.: Encoding raises least-squares F1 from 0.4095 on raw inputs to 0.4301 (Table I); logistic GD reaches 0.4390, the best linear result. The tested L_2 penalty changes logistic F1 by only −0.0002. GD, SGD, and direct least squares fit the same MSE model, so their differences partly reflect finite optimization budgets.

Figure 1 shows an exploratory rise from 0.4172 to 0.4400 F1 after recoding, missingness flags, and one-hot encoding. Because several changes were combined, it does not isolate their effects. A separate five-fold check found only −0.0003 F1 from numeric cleaning once one-hot encoding was present.

Model choice and refinements.: The five-fold baseline table and the 15-fold model comparison use different threshold procedures, so their F1 values should be read within each table. On the 15 shared folds, the selected three-seed GBDT improves F1 over logistic regression by 0.0077 (corrected 95% interval $[0.0038, 0.0115]$) [6]. The test submissions score 0.449 F1 for logistic regression and 0.454 for this GBDT.

The MLP achieves 0.4399, close to logistic regression. The feature bag uses answer counts and target-ordered codes, reaching 0.4474, but its paired gain over the selected GBDT is only 0.0011 (corrected $t = 1.51$). Other refinements in Table II show similarly small gains. We retain the three-seed GBDT for its documented submission and reproducible recipe. Selection used the same folds, so these scores are not fully nested estimates of the complete model search.

Ordered-code splits reduce best-stage log-loss from 0.21474 to 0.21403 (Fig. 2). At rate 0.05, 550 trees lie in a flat 500–900-tree region; the default overfits at 2,000 trees (F1 0.4323). Rate 0.02 gives no clear F1 gain. Best-stage F1 optimizes thresholds on the evaluated folds and is optimistic.

IV. CONCLUSIONS

Encoding supports a strong logistic baseline; GBDT provides the main gain, while further refinements remain uncertain. The final `run.py` refits three GBDTs on all labeled rows, averages logits, and applies threshold 0.2166; its documented AIcrowd F1 is 0.454.

All 773 training respondents answering `HAREHAB1`, a question about rehabilitation after an infarction, are positive. This post-diagnosis signal limits interpretation as prospective clinical risk prediction.

References

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